BNPL Credit Risk — Modeling (Logistic Regression and k-NN)

ISOM3018 Course Project

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Section 1: Modeling Overview

Two classification models — logistic regression and k-nearest neighbors (k-NN) — are built to predict whether a BNPL customer will default on their repayment (`default_flag`: Default vs. No Default).

Both models are trained on the same preprocessed training set (SMOTE applied to address class imbalance) and evaluated on a held-out test set. Numeric predictors are standardized before fitting. The positive class for all discrimination metrics is **Default**.

Section 2: Logistic Regression

Logistic regression models the probability of default as a linear function of customer characteristics. It is selected as a baseline for its interpretability — the contribution of each predictor can be directly examined. All numeric predictors are standardized; categorical predictors are dummy-encoded.

Table 1: Test Set Performance — Logistic Regression

Metric	Value
roc_auc	0.7449
accuracy	0.6617
sens	0.6584
spec	0.6638

Interpretation: The model achieves a ROC AUC of 0.745, indicating moderate discriminative ability. At the default 0.50 threshold, it correctly identifies 65.8% of actual defaulters (sensitivity), while maintaining a similar rate for non-defaulters (specificity = 0.664). For a BNPL provider, higher sensitivity is generally preferred — missing a defaulter is more costly than a false alarm.

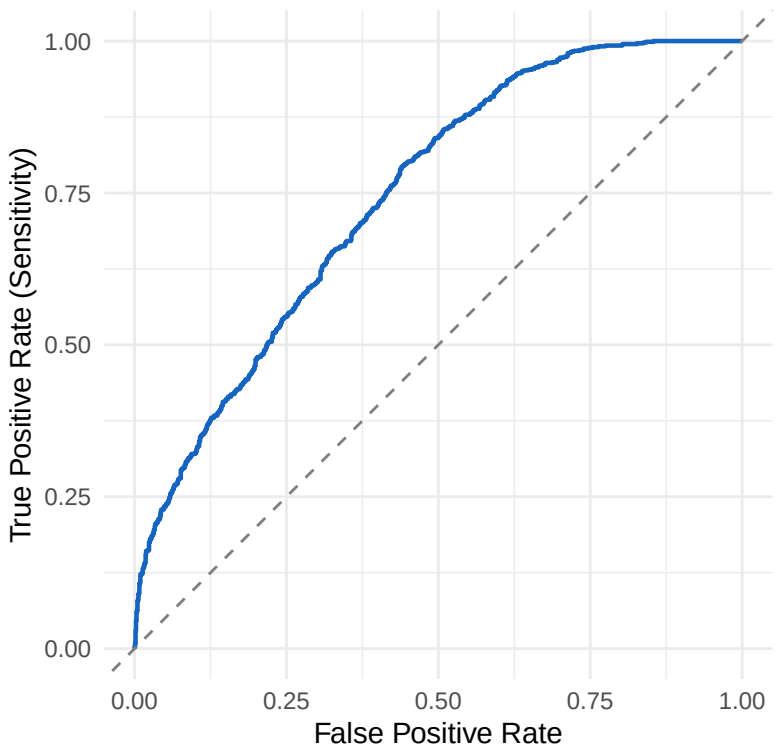

Figure 1: ROC Curve — Logistic Regression (Test Set).

Interpretation: The ROC curve lies clearly above the diagonal (random baseline), confirming the model has meaningful predictive power for identifying BNPL default risk.

Table 2: Confusion Matrix — Logistic Regression (Test Set)

	Actual: Default	Actual: No Default

Predicted: Default	532	424
Predicted: No Default	276	837

Interpretation: Of the 808 customers who actually defaulted, 532 were correctly flagged (true positives), while 276 were missed (false negatives). Of the 1,261 non-defaulters, 837 were correctly cleared and 424 were incorrectly flagged. Reducing the false negative count — undetected defaulters — is the primary operational priority for credit risk management.

Section 3: k-Nearest Neighbors (k-NN)

k-NN classifies each customer based on the majority default outcome among their k nearest neighbors in standardized feature space. The optimal k is selected by 5-fold cross-validation on the training data, choosing the value that maximizes ROC AUC.

Table 3: Optimal k Selected by Cross-validation

Selected k	Selection Criterion
43	Max CV ROC AUC

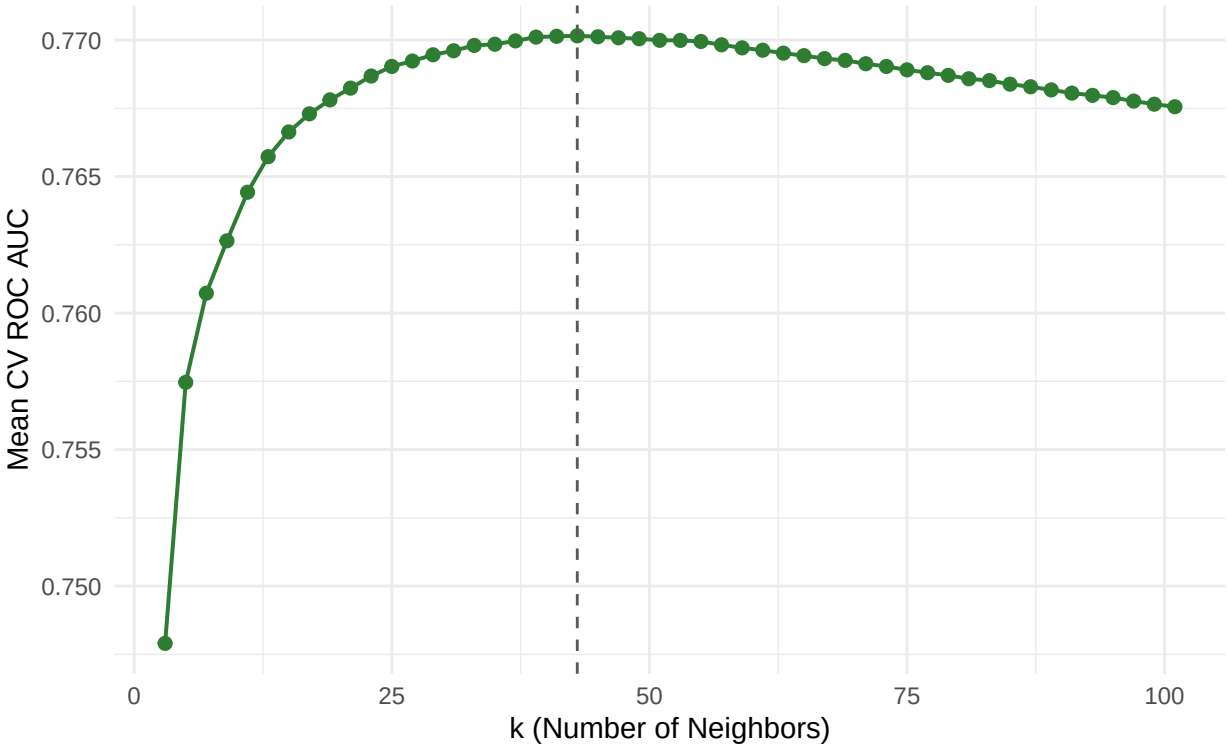

Figure 2: Cross-validated ROC AUC by k (Training Data Only).

Interpretation: ROC AUC increases steadily as k grows, peaking at k = 43. A larger k produces smoother, more stable predictions by averaging over more neighbors, which helps on this dataset.

Table 4: Test Set Performance — k-NN

Metric	Value
roc_auc	0.7137
accuracy	0.6593
sens	0.6163
spec	0.6868

Interpretation: k-NN achieves a ROC AUC of 0.714, slightly below logistic regression. Its sensitivity (true positive rate for defaulters) is 0.616, meaning it detects fewer defaulters compared to logistic regression. However, it achieves a higher specificity, correctly clearing more non-defaulters.

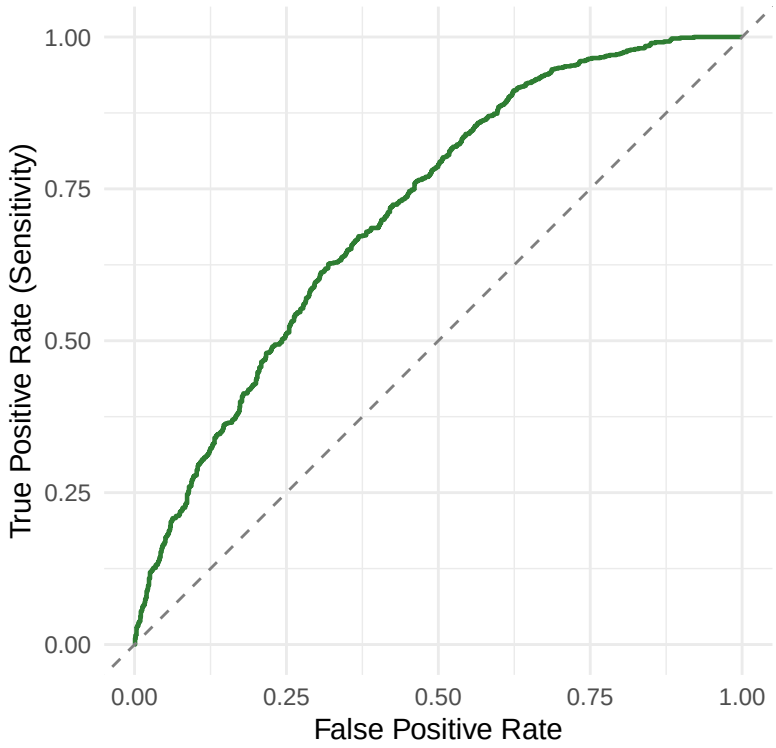

Figure 3: ROC Curve — k-NN (Test Set).

Table 5: Confusion Matrix — k-NN (Test Set)

	Actual: Default	Actual: No Default
Predicted: Default	498	395
Predicted: No Default	310	866

Interpretation: k-NN correctly identifies 498 defaulters (true positives) but misses 310 (false negatives). Compared to logistic regression, it catches fewer actual defaulters — a disadvantage in credit risk contexts where undetected defaults carry higher financial cost.

Section 4: Model Comparison

Table 6: Test Set Performance — Logistic Regression vs. k-NN

Model	ROC AUC	Accuracy	Sensitivity	Specificity
Logistic Regression	0.7449	0.6617	0.6584	0.6638
k-NN	0.7137	0.6593	0.6163	0.6868

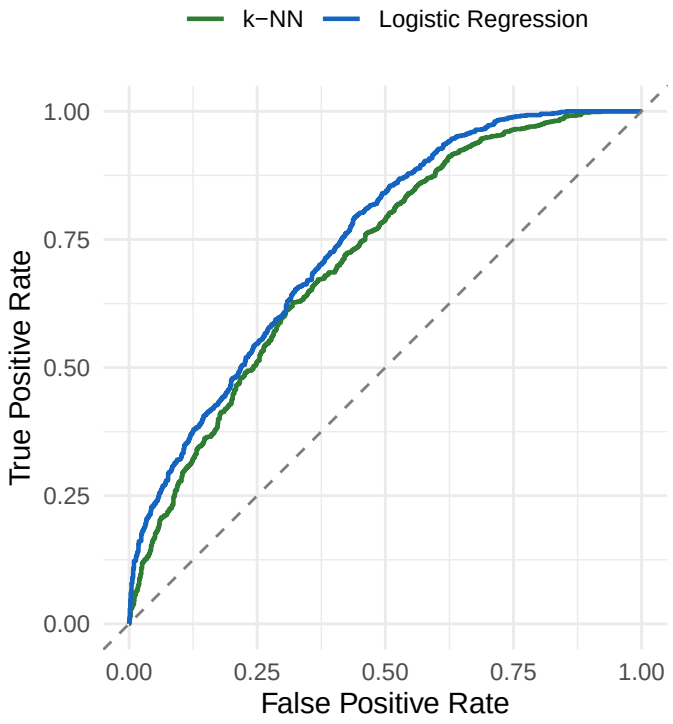

Figure 4: ROC Curves — Logistic Regression vs. k-NN (Test Set).

Interpretation: Logistic regression outperforms k-NN on ROC AUC (0.745 vs. 0.714), sensitivity, and accuracy. k-NN achieves a slightly higher specificity (0.687 vs. 0.664). For default detection where catching defaulters is the priority, logistic regression is the stronger performer of the two.

Section 5: Business Implications

- **Logistic regression** is the better-performing model of the two and has the added advantage of interpretability — each predictor's effect on default probability can be quantified and explained to risk managers and regulators.
- **k-NN** performs slightly below logistic regression on most metrics. Its lower sensitivity means more defaults go undetected, which is a significant drawback in a credit risk context.
- At the current 0.50 probability threshold, both models miss roughly a third of actual defaulters. A BNPL provider could lower the threshold to flag more potential defaulters, trading some specificity for higher sensitivity — a business decision depending on the relative cost of missed defaults versus false alarms.
- The predicted default probabilities from both models can support practical risk actions: prioritizing manual review for borderline cases, applying lower credit limits to high-risk customers, and triggering early repayment reminders.